

Deep Equilibrium Models

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Abstract

We present a new approach to modeling sequential data: the deep equilibrium model (DEQ). Motivated by an observation that the hidden layers of many existing deep sequence models converge towards some fixed point, we propose the DEQ approach that *directly* finds these equilibrium points via root-finding. Such a method is equivalent to running an *infinite* depth (weight-tied) feedforward network, but has the notable advantage that we can analytically backpropagate through the equilibrium point using implicit differentiation. Using this approach, training and prediction in these networks require only *constant* memory, regardless of the effective “depth” of the network. We demonstrate how DEQs can be applied to two state-of-the-art deep sequence models: self-attention transformers and trellis networks. On large-scale language modeling tasks, such as the WikiText-103 benchmark, we show that DEQs 1) often improve performance over these state-of-the-art models (for similar parameter counts); 2) have similar computational requirements to existing models; and 3) vastly reduce memory consumption (often the bottleneck for training large sequence models), demonstrating an up-to 88% memory reduction in our experiments. The code is available at <https://github.com/locuslab/deq>.

1 Introduction

Most modern feedforward deep networks are built on the core concept of *layers*. In the forward pass, each network consists of a stack of some L transformations, where L is the depth of the network. To update these networks, the backward passes rely on backpropagating through the same L layers via the chain rule, which typically necessitates that we store the intermediate values of these layers. The value for L is usually a hyperparameter and is picked by model designers (e.g., ResNet-101 [25]). Among the many applications of deep networks, sequence modeling has witnessed continuous advances in model architectures. Specifically, while recurrent networks have long been the dominant model for sequences [21, 26, 14, 34], deep feedforward architectures based on temporal convolutions [49, 47, 7] and self-attention [48, 16, 13] have (re-)emerged to claim superior performance on a variety of sequence prediction tasks.

In very general terms, a deep feedforward sequence model can be written as the following iteration:

$$\mathbf{z}_{1:T}^{[i+1]} = f_{\theta}^{[i]}(\mathbf{z}_{1:T}^{[i]}; \mathbf{x}_{1:T}) \quad \text{for } i = 0, 1, 2, \dots, L-1 \quad (1)$$

where i is the layer index; $\mathbf{z}_{1:T}^{[i]}$ is the hidden sequence of length T at layer i ; $\mathbf{x}_{1:T}$ is the input sequence (i.e., we are choosing to explicitly model skip connections, for reasons we explain later); and $f_{\theta}^{[i]}$ is some nonlinear transformation which typically enforces causality (i.e., future time points cannot influence past ones). Our paper derives its motivation from surprising recent works that employ the *same* transformation in each layer (known as *weight tying*, with $f_{\theta}^{[i]} = f_{\theta}, \forall i$) and still achieve results competitive with the state-of-the-art [18, 8, 15]. This raises an interesting question: If the same transformation is applied at each layer of a deep network, what is the limit of this process, and how do we model it?

In this paper, we propose a new approach to “deep” modeling that addresses this question. Specifically, we introduce the deep equilibrium model (DEQ), a method that directly computes the fixed point $\mathbf{z}_{1:T}^*$ of a nonlinear transformation, i.e., the solution to the nonlinear system

$$\mathbf{z}_{1:T}^* = f_{\theta}(\mathbf{z}_{1:T}^*; \mathbf{x}_{1:T}). \quad (2)$$

This solution corresponds to the eventual hidden layer values of an *infinite depth* network. But instead of finding this value by iterating the model, we propose to directly (and in practice, more quickly) solve for the equilibrium via any black-box root-finding method. Importantly, we show that DEQ can *directly* differentiate through the fixed point equations via implicit differentiation, which does not require storing *any* intermediate activation values. In other words, we can backpropagate through the infinite-depth network while using only *constant* memory, equivalent to a single layer’s activations.

After developing the generic DEQ approach, we study in detail the instantiation of DEQ via two feedforward sequence models: *trellis networks* (weight-tied temporal convolutions) [8] and memory-augmented *universal transformers* (weight-tied multi-head self-attention) [18, 16], both of which have obtained state-of-the-art performance (SOTA) on various sequence tasks. We show how both the forward and backward passes can be implemented efficiently via quasi-Newton methods. Finally, we demonstrate via experiments on large-scale high-dimensional sequence modeling benchmarks (e.g., WikiText-103 language modeling) that, despite only using constant memory, DEQ can attain modeling accuracy on par with (or even slightly better than) corresponding layer-based networks. We believe that DEQ offers a novel perspective on the analysis of sequential data.

2 Background

Deep sequence models. Given an input sequence $\mathbf{x}_{1:T} = [x_1, \dots, x_T] \in \mathbb{R}^{T \times p}$, where $x_i \in \mathbb{R}^p$ (e.g., a word embedding) and $T \in \mathbb{N}$ is the sequence length, we define a sequence model as any function G that produces output $G(\mathbf{x}_{1:T}) = \mathbf{y}_{1:T} \in \mathbb{R}^{T \times q}$ that satisfies the causality constraint: y_t depends only on $\mathbf{x}_{1:t}$ and not on any element of $\mathbf{x}_{t+1:T}$. Recent progress on autoregressive sequence tasks has been based on deep learning, where three major families of sequence models stand out. Recurrent networks (RNNs) [21, 51] as well as their variants such as LSTM [26] are universally applied and optimized in a variety of time-series tasks [9, 22, 34]. Alternatively, prior work has shown that deeply stacked temporal convolutions [49, 47, 17, 7] can achieve competitive results, especially on long sequences. Finally, the self-attention transformer architecture [48, 16] has also achieved SOTA on several NLP benchmarks [19, 13]. Efforts have also been devoted to drawing deeper connections among the three model families. Bai et al. [8] study the underlying relationship between RNNs and ConvNets, unifying these in the Trellis Network, which combines the benefits of both families. Dehghani et al. [18] introduce a recurrently-stacked universal transformer and demonstrate its effectiveness on text understanding and generation.

Memory-efficient deep networks. An important factor that limits the development of high-capacity networks is limited memory on hardware devices used for training. To address this issue, [12] proposes gradient checkpointing that reduces an L -layer network’s memory requirement to $O(\sqrt{L})$ at the cost of extra forward passes (i.e., extra computations). Alternatively, [23, 30] develop reversible networks, where each layer’s activations can be reconstructed from the next layer during backpropagation to reduce memory requirements. DEQs reduce memory consumption to a *constant* (i.e., independent of network “depth”) by directly differentiating through the equilibrium point and thus circumventing the construction and maintenance of “layers”.

Continuous view of deep networks. Some prior works have studied continuous views of deep networks. [41] proposes a biologically inspired equilibrium propagation framework for an energy-based model whose prediction is the fixed-point of the energy dynamics at its local minimum. [24, 11] model deep ResNets by black-box ODE solvers in forward and backward passes (as if the network has smaller “layer steps”) given the start- and end-points of a dynamical system. For deep sequence models, [43, 36] consider the RNN as a dynamical system to investigate its stability properties.

Our work takes a further step in the direction of the aforementioned areas. While some of the prior work has primarily focused on the analysis of residual architectures or small symmetric-weight energy-based models, our work is not predicated on any specific type of interlayer transformation. We show that DEQs can be easily instantiated via two very different sequence learning architectures. More fundamentally, unlike ODE-based methods, which use the adjoint system to backpropagate

through the entire latent trajectory, the DEQ model solves directly for sequence-level equilibria via a quasi-Newton method and backpropagates directly through this fixed point, without regard for the solution path that brought it there. Moreover, while ODE-based models [24, 11] were verified on numerical experiments and MNIST classification, computation and numerical stability issues challenge their application to large-scale problems. In comparison, we demonstrate the applicability of DEQs on realistic high-dimensional sequence tasks with competitive performance, while enjoying similar constant-memory benefits as [11].

Implicit layers in deep learning. The DEQ model can be viewed as an infinitely deep network, but interestingly can also be viewed as a *single*-layer network, with the caveat that the layer is defined *implicitly*: the output $\mathbf{z}_{1:T}^*$ is defined as the value which solves some non-linear equation. There has been a growing interest in implicit layers in recent years [37, 3, 37, 50], but the precise formulation of the DEQ is quite different, and our current models represent the largest-scale practical application of implicit layers in deep learning of which we are aware. Concurrent work [20] also looks at such implicit layers in a broad sense and focuses on training small models via Lagrangian methods; a combination of these approaches with the DEQ model is a promising avenue for future work.

Another thread of work on implicit layers traces back to some of the original papers on recurrent networks trained via recurrent backpropagation (RBP) [2, 38]. Recent work [28] has re-examined RBP and established an implicit, constant-memory variant based on conjugate gradient and Neumann series. A number of related papers also enforce fixed point conditions within RNN architectures [54, 27]. Whereas the DEQ model shares similarities with the RBP approach, some major differences involve: 1) the explicit use of equilibrium as a replacement for depth in general networks, along with our proof of the universality of these models to replace depth; 2) the use of the approach in methods outside of fixed-input RNNs (i.e., same input vector x_t for all t), especially the compatibility with SOTA architectures; and 3) the scalability of the DEQ model to practical tasks where it achieves results on par with the current SOTA, whereas RBP has typically been applied in small-scale settings.

3 The Deep Equilibrium Sequence Model

We broadly consider the class of *weight-tied* deep sequence models (with passthrough connections from the input to each layer), which consist of the update

$$\mathbf{z}_{1:T}^{[i+1]} = f_{\theta}(\mathbf{z}_{1:T}^{[i]}; \mathbf{x}_{1:T}), \quad i = 0, \dots, L-1, \quad \mathbf{z}_{1:T}^{[0]} = \mathbf{0}, \quad G(\mathbf{x}_{1:T}) \equiv \mathbf{z}_{1:T}^{[L]} \quad (3)$$

We note that this model encapsulates classes such as the trellis network [8] and the universal transformer [18] (which is typically not written with passthrough connections, but this is a trivial modification). Such weight-tying is generally considered to come with four major benefits: 1) it acts as a form of regularization that stabilizes training and supports generalization; 2) it significantly reduces the model size; 3) it is trivial to show that *any* deep network can be represented by a weight-tied deep network of equal depth and only a linear increase in width (see Appendix C); and 4) the network can be unrolled to *any* depth, typically with improved feature abstractions as depth increases [8, 18]. However, in practice almost all such models (and deep nets in general) are stacked, trained and evaluated *by unrolling a pre-determined, fixed number of layers*. One reason is the limited memory on training hardware: the models need to store intermediate hidden units for backpropagation and thus cannot be trained beyond a certain depth that depends on the available memory.

In principle, the network could have infinite depth. This is attained in the limit of unrolling a weight-tied model for an ever higher number of layers. What is the limit of this process? In practice, for certain classes of f_{θ} (discussed later), we hypothesize and observe that such weight-tied models tend to converge to a *fixed point* as depth increases towards infinity (see Appendix D for empirical evidence). In other words, as each layer refines the previous one by combining temporal features across the sequence, increasing depth towards infinity brings “diminishing returns”: each additional layer has a smaller and smaller contribution until the network reaches an equilibrium:

$$\lim_{i \rightarrow \infty} \mathbf{z}_{1:T}^{[i]} = \lim_{i \rightarrow \infty} f_{\theta}(\mathbf{z}_{1:T}^{[i]}; \mathbf{x}_{1:T}) \equiv f_{\theta}(\mathbf{z}_{1:T}^*; \mathbf{x}_{1:T}) = \mathbf{z}_{1:T}^* \quad (4)$$

3.1 The Deep Equilibrium Approach

We introduce the deep equilibrium model (DEQ) which, instead of iteratively stacking f_{θ} , directly solves for and differentiates through the equilibrium state.